

MingYu Zhao

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EDUCATION

Rutgers University, Graduate School of Arts and Sciences

Ph.D. in Computer Science

University of Science and Technology of China

B.Eng. in Artificial Intelligence

New Brunswick, NJ

Expected May 2029

Hefei, China

Jun 2024

PUBLICATIONS

“” denotes equal contribution.*

First-author Publications

- **Mingyu Zhao***, Y. Zhou*, Z. Wang, D. Gu, B. Guo, R. Ye, L. Han, C. Jin, D. N. Metaxas. “M3-Bench: Multi-Modal, Multi-Hop, Multi-Threaded Tool-Using MLLM Agent Benchmark.” Submitted to the *NeurIPS*, 2026.
- **Mingyu Zhao**, Z. Yang, Y. Zhou, Z. Xia, C. Jin, X. He, C. Neidle, D. N. Metaxas. “MHB: Multimodal Handshape-aware Boundary Detection for Continuous Sign Language Recognition.” Accepted by the *LREC*, 2026.
- **Mingyu Zhao***, Z. Li*, X. Dong, H. Ling, B. Huang. “CAPAA: Classifier-Agnostic Projector-Based Adversarial Attack.” Accepted by the *IEEE ICME*, 2025.

Selected Other Publications

- “Lor-VP: Low-rank Visual Prompting for Efficient Vision Model Adaptation.” *ICLR 2025*.
- “Large Sign Language Models: Toward 3D American Sign Language Translation.” *WACV 2026*.
- “LED: LLM-Enhanced Open-Vocabulary Object Detection without Human-Curated Data Generation.” *CVPR 2026*.

RESEARCH EXPERIENCE

Topic: Multi-Hop, Multi-Threaded MLLM Agent Evaluation and Optimization *Sep. 2025 ~ Apr. 2026*

Instructor: Prof. [Dimitris N. Metaxas](#) (Rutgers University)

- Built an end-to-end MCP benchmark generation and correctness validation framework; designed an MCP benchmark suite with 200+ tasks involving visual grounding and cross-tool dependencies, each with a standardized trajectory.
- Proposed a structure-aware tool-call evaluation metric using **sentence-encoder** embeddings and **similarity-bucketed Hungarian matching**; quantitatively evaluated GPT, Gemini, Claude, and **10+** MLLM models across the benchmark, with qualitative and quantitative analyses of budget and error types to characterize tool-call failure modes.
- Performed **supervised fine-tuning** on LLaDA 1.5 using generated tool-call data from multiple datasets, integrated a reasoning module, and optimized the model with **reinforcement learning**, improving specified tool-call task performance by **32%** over the base LLaDA 1.5 tool-calling baseline under **LLM judge** scores and accuracy metrics.

Topic: Multi-Modal Video Temporal Boundary Detection

Jan. 2025 ~ Dec. 2025

Instructor: Prof. [Dimitris N. Metaxas](#) (Rutgers University) and Prof. [Carol Neidle](#) (Boston University)

- Designed a spatio-temporal convolution-based segmentation module with velocity and acceleration features from skeleton sequences, improving **temporal boundary detection** F1 score by around **10%** over previous methods.
- Integrated handshape information by pretraining a handshape classifier over canonical categories and fusing the classifier’s learned hidden representations into the segmentation stream via **cross-attention** with a gating mechanism.
- Built an **end-to-end** continuous sign language recognition pipeline by combining boundary detection with sign recognition, improving boundary detection F1 score by nearly **20%** overall; for signs with well-localized boundaries, continuous recognition performance was within only **10%** of isolated-sign recognition.

Topic: Classifier-Agnostic Projector-Based Adversarial Attack

July. 2024 ~ Dec. 2024

Instructor: Prof. [HaiBin Ling](#) (Stony Brook University) and Prof. [BingYao Huang](#) (Southwest University)

- Proposed a pose-agnostic projector-based attack algorithm using Grad-CAM++ to focus perturbations on salient physical regions, mitigating the effects of camera-pose changes and improving visual stealthiness by around **20%**.
- Designed a novel optimal loss function to generate projector-based adversarial attacks across multiple classifiers with **negligible performance degradation** compared with classifier-specific attacks.
- Validated the effectiveness of the proposed CAPAA by building a camera-projector system in the real world and by evaluating the attack success rate and stealthiness on 10 sets of ablation experiments.

INDUSTRY PROJECT

Topic: ASL Pal

Sep. 2025 ~ Present

Leads: [Beryl Breindel](#) (DawnSignPress) and Prof. [Dimitris N. Metaxas](#) (Rutgers University)

- Fine-tuned a state-of-the-art isolated American Sign Language (ASL) recognition model on the company’s proprietary ASL dataset, achieving **85%** recognition accuracy and outperforming the original model.
- Engineered the backend inference pipeline with model compression, parallel execution, model preloading, and other techniques, reducing model runtime from **5 min** to **15 sec**.
- Integrated the models into the web platform with the engineering team and validated the system through user testing.

TECHNICAL STRENGTHS

Skills: Python, PyTorch, Bash, C/C++, AI-assisted IDE, Agentic Coding, Linux, MCP, Git, Flask, Slurm, L^AT_EX.